Department of Computer Science and Engineering

University of Chittagong

8th Semester B.Sc. Engineering Exam 2024

CSE 815: Machine Learning

Marks: 54 Time: 4 hours

[Answer any three of the four questions from each of the two sections below. Answers to parts of a question should appear contiguously in the answer script. Ensure readability of your handwriting.]

Section A

A-1| (a) Define the supervised classification task in terms of a mapping function from an input space $\mathcal{X}$ to a discrete label space $\mathcal{Y}$. Given a loss function $L(f(x), y)$, explain how optimization minimizes empirical risk over a given dataset. (3)

(c) Consider a simple linear regression model $y = wx + b$. Derive the closed-form solution for w and b that minimizes the mean squared error on the training data. (3)

(d) Use the diagram below to illustrate the geometric interpretation of error minimization in linear regression. Label the hypothesis line and sample points. (3)

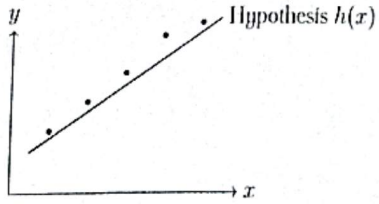


Figure 1: Question A-1(d): Linear Regression Hypothesis

A-2| (a) What is a support vector? Why is the maximum margin in the SVM technique? How do you compute M in terms of W and b? (2)

(b) Write down the Q-learning algorithm. (2)

(c) Define convolution in the context of CNNs. Explain the drawbacks of fully connected Deep Neural Networks that motivated the development of CNNs. (2)

(d) An input volume of size $32 \times 32 \times 3$ is convolved with 10 filters of size $5 \times 5 \times 3$, stride $S = 1$, and zero padding $P = 2$. Calculate the output volume size and the total number of trainable parameters in this layer. (3)

A-3| (a) Describe the Hidden Markov Models (HMM) with a necessary diagram. (4)

(b) Consider an HMM with an explicit absorber state and unique null visible symbol v_0 with the following transition probabilities a_{ij} and symbol probabilities b_{jk} (where the matrix indexes begin at 0):

$$a_{ij} = \begin{array}{c} \\ v_0 \\ v_1 \\ v_2 \\ v_3 \end{array} \begin{array}{c} \begin{array}{cccc} w_0 & w_1 & w_2 & w_3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0.2 & 0.3 & 0.1 & 0.4 \\ 0.2 & 0.5 & 0.2 & 0.1 \\ 0.8 & 0.1 & 0.0 & 0.1 \end{pmatrix} \end{array} \qquad b_{jk} = \begin{array}{c} \\ w_0 \\ w_1 \\ w_2 \\ w_3 \end{array} \begin{array}{c} \begin{array}{ccccc} v_0 & v_1 & v_2 & v_3 & v_4 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0.3 & 0.4 & 0.1 & 0.2 \\ 0 & 0.1 & 0.1 & 0.7 & 0.1 \\ 0 & 0.5 & 0.2 & 0.1 & 0.2 \end{pmatrix} \end{array}$$

i. Give a graph representation of this Hidden Markov Model. (1)

ii. Suppose the initial hidden state at $t = 0$ is w_1. Starting from $t = 1$, what is the probability it generates the particular sequence $V^4 = \{v_1, v_3, v_2, v_0\}$? (2)

iii. Given the above sequence V^4, what is the most probable sequence of hidden states? (2)

A-4| (a) What is the purpose of a pooling layer in a CNN? Differentiate between max pooling and average pooling. (2)

(b) A 7×7 input feature map is passed through a 2×2 max pooling filter with stride $S = 2$ and no padding. Calculate the output spatial size. (1.5)

(c) Define the ReLU activation function mathematically. Why is ReLU preferred over sigmoid in deep CNNs? (1.5)

(d) Given the following input matrix and filter, perform one convolution operation (stride = 1, no padding).

Input: $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ **Filter:** $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Calculate the resulting feature map. (1.5)

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(c) Explain parameter sharing (e.g., weight sharing and pooling). How does it contribute to translation invariance? Use an example of a shifted object in an image. (2.5)

SECTION B

B-1/ (a) What is overfitting? Explain the consequences of overfitting and respective handling techniques with necessary examples? (3)

(b) Why is data called the "fuel" for AI? Justify with two examples. Identify and explain any two Data Challenges in AI implementation. (3)

(c) Describe a situation (in the context of classification) where using KNN would be more reasonable than using logistic regression.(3)

B-2/ (a) Given a binary classifier, define the following metrics formally: Precision, Recall, Accuracy, F1-score. Provide mathematical formulas for each and explain in which situation each is most informative. (3)

(b) Given the following confusion matrix for a binary classification problem, calculate precision, recall, accuracy, and F1-score. (3)

		Predicted	
		Positive	Negative
Actual	Positive	40	10
	Negative	5	45

Provide all necessary calculations and intermediate steps.

(c) Sketch and explain a ROC curve. Indicate where a perfect classifier, a random guesser, and a poor classifier would lie. Then, define the AUC metric and explain its relevance to model comparison. (3)

B-3/ (a) Define Decision Tree and Entropy. (3)

(b) Given the Tennis playing record as shown in the following Table. Draw the Decision Tree for the respective Tennis playing record. You have to show a detailed step-by-step procedure for drawing the decision tree. (6)

Day	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

B-4/ (a) Write down the KNN algorithm. What are the basic similarities and dissimilarities between KNN and K-means clustering methods? (4)

(b) Suppose we have height and weight and its corresponding T-shirt size of several customers. Your task is to predict the T-shirt size of Mr. Perfect, whose height is 161 cm and his weight is 61 kg. (5)

Data including height, weight, and T-shirt size information are (Height in cms, Weight in kgs, T-shirt Size):
(158, 58, M), (158, 59, M), (158, 63, M), (160, 59, M), (160, 60, M), (163, 60, M), (163, 61, M), (160, 64, L), (163, 64, L), (165, 61, L), (165, 62, L), (165, 65, L), (168, 62, L), (168, 63, L), (168, 66, L), (170, 63, L), (170, 64, L), (170, 68, L).

Solve the above problem using KNN algorithm with $k = 3$ or 5. You may use either Euclidean or Manhattan as a distance function.

Euclidean: $d(x,y) = \sqrt{\sum_{i=1}^{m}(x_i - y_i)^2}$ Manhattan: $d(x,y) = \sum_{i=1}^{m}|x_i - y_i|$

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University of Chittagong
Department of Computer Science and Engineering
8th Semester B.Sc Engineering Examination 2022
Course Code: CSE-815 Course Title: Machine Learning
Full Marks: 54 Time: 4hours

[Answer any 3 questions from each section. Figures in the right hand margin indicate full marks]

Section A:

1 a) What are the differences between Artificial Intelligence (AI) vs Machine Learning (ML) vs Deep Learning (DL)? 2

b) What are the steps for solving any problem using machine learning? 3

c) Answer the provided questions using the example confusion matrix, which represents a general land cover classification. 4

		Reference			
		Developed	Low Vegetation	Forest	Water
Classification	Developed	152	35	11	13
	Low Vegetation	31	189	17	5
	Forest	2	27	231	11
	Water	7	6	8	93

i) How many total validation samples were collected?

ii) How many samples were correct?

iii) What was the overall accuracy of the classification?

iv) Measure Precision, Recall and F-1 score.

2 a) Suppose you are given the following pairs. You will simulate the k-means algorithm to identify TWO clusters in the data. 5

Data #	x	y
1	1.90	0.97
2	1.76	0.84
3	2.32	1.63
4	2.31	2.09
5	1.14	2.11
6	5.02	3.02
7	5.74	3.84
8	2.25	3.47
9	4.71	3.60
10	3.17	4.96

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Suppose you are given initial assignment cluster center as {cluster1: #1}, {cluster2: #10} – the first data point is used as the first cluster center and the 10-th as the second cluster center. Please simulate the k-means (k=2) algorithm for ONE iteration. Identify the cluster assignments after one iteration? Assume k-means uses Euclidean distance.

b) Explain the Agglomerative Hierarchical Clustering algorithm with the help of an example.

c) What are some stopping criteria for K-means clustering?

3 a) You have data from the questionnaires survey and objective testing with two attributes (acid durability and strength) to classify whether a special paper tissue is good or not. Here are four training samples.

X1= Acid Durability (seconds)	X2= Strength (kg/ square meter)	Classification
7	7	Bad
7	4	Bad
3	4	Good
1	4	Good

Now the factory produces a new paper tissue that pass laboratory test with X1= 3 and X2= 7. Apply KNN algorithm to guess what the classification of this new tissue is?

b) Compare between K-means and KNN algorithms.

c) What is AUC ROC curve? What does a ROC curve of 0.8 mean?

4 a) How does the Naive Bayes classification method perform in scenarios with correlated features? Can its assumption of feature independence compromise its accuracy in real-world datasets? What strategies can be employed to mitigate the impact of this assumption on classification outcomes?

b) In what ways does Naive Bayes classification focuses on learning the conditional probability of classes given features resemble the principles of discriminative classifiers? Can Naive Bayes be considered a hybrid approach, drawing elements from both generative and discriminative paradigms, or does it predominantly adhere to one category over the other?

c) Consider the following data set for Naive Bayesian classification problem.

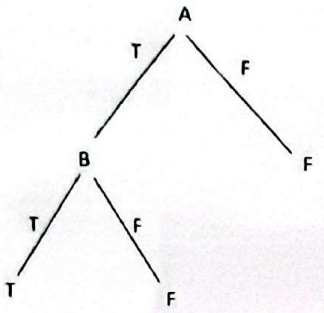
Age	Income	Credit_Rating	Loan (Class)
High	Fair	Low	Risk
Low	Fair	High	No Risk
Low	Excellent	High	No Risk

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High	Fair	High	Risk

Utilize naive Bayesian classification to forecast the class designation of a test sample (with options A=low, B=Fair, C=Low). Consider employing Laplacian correction to prevent zero probability computations if desired.

Section B:

a)	Is the following figure represents the decision tree for OR function? Explain your answer.  	1
b)	Consider the problem of predictiong whether the university will be closed on a particular factors which decide this are whether there is a natural disasters, whether it is a weekend or an official holiday. Suppose we have the training examples describe in following table.	4

Disaster	Holiday	Weekend	Closed
T	T	F	F
T	T	F	T
F	T	F	F
T	T	F	F
F	F	F	F
F	F	F	T
T	F	F	T
F	F	F	T

i) What would be the effect of the Weekend attribute on the decision tree if it were made the root? Explain in terms of information gain.

ii) If you cannot make Weekend the root node, which attribute should be made the root node of the decision tree? Explain your reasoning and show your calculations. (You may use $\log_2 0.75 = -0.4$ and $\log_2 0.25 = -2$)

c)	*Consider the dataset given below where T and F represent True and False respectively. What is the entropy H (Rain)?*	4

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Temperature	Cloud	Rain
Low	T	T
Low	T	T
Medium	T	F
Medium	T	T
High	T	F
High	F	F

6 How does the kernel trick enhance the effectiveness of Support Vector Machines (SVMs) in handling nonlinear classification tasks? Compare and contrast the performance of linear SVMs versus kernelized SVMs on datasets with varying degrees of complexity.

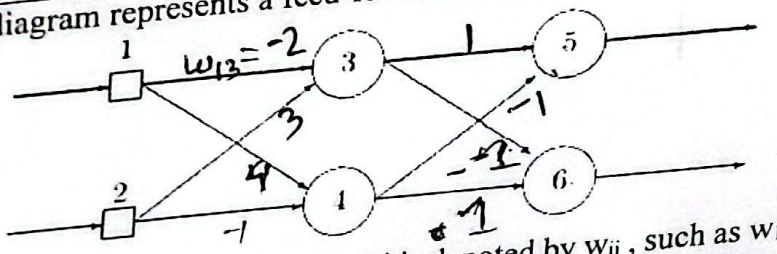
Discuss the key principles behind margin maximization and the selection of support vectors in SVMs, highlighting their role in achieving robust classification boundaries. Investigate the impact of different kernel functions (e.g., polynomial, radial basis function) on the decision boundary flexibility and generalization capability of SVMs.

For instance, we are given the following data points in 2D space:

x	1	4	1	2	5	5	2	6
y	1	0	-1	1	1	-1	-1	0

Classify all data points either positive or negative when $w = \{2,0\}$ and $b=-2$. What would be the distance from the separating hyperplane to any point on either positive or negative class? Finally, find the maximum margin.

7 The following diagram represents a feed-forward neural network with one hidden layer:



A weight on connection between nodes i and j is denoted by w_{ij}, such as w_{13} is the weight on the connection between nodes 1 and 3. The following table lists all the weights in the network:

$w_{13} = -2$ $w_{35} = 1$ $w_{23} = 3$ $w_{45} = -1$ $w_{14} = 4$ $w_{36} = -1$ $w_{24} = -1$ $w_{46} = 1$

Each of the nodes 3, 4, 5 and 6 uses the following activation function:

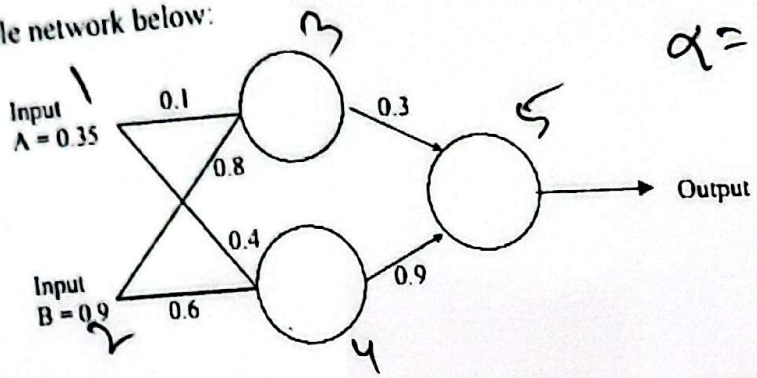
$$g(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Where y denotes the weighted sum of a node. Each of the input nodes (1 and 2) can only receive binary values (either 0 or 1). Calculate the output of the network (y_5 and y_6) for each of the input patterns:

Pattern:	P_1	P_2	P_3	P_4
Node 1:	0	1	0	1
Node 2:	0	0	1	1

5

Consider the simple network below:



Assume that the neurons have a Sigmoid activation function and

(i) Perform a forward pass on the network.
(ii) Perform a reverse pass (training) once (target = 0.5).
(iii) Perform a further forward pass and comment on the result.

8 a) Explain the architecture of a convolutional neural network (CNN).

b) Consider that your input images have size 227×227×3 and filter of size 11×11×3 in the first convolution layer. In the case of AlexNet we have in the first convolution layer a total of 96 filters, a stride of 4 and a padding of 0.

i. Determine the width/height of the output of this first convolution layer.
ii. Determine the number of units within the first convolutional layer.
iii. Determine the number of trainable parameters/weights within the convolutional layer with weight sharing
iv. How many parameters would we have if we had used a FF layer with 256 units instead of a convolutional layer?

c) Differentiate between Federated learning and Incremental learning.

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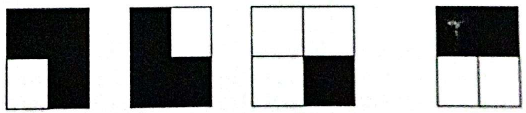
[Answer the six of the following questions Three from each Section A and Section B; Figures in the right-hand margin indicate full marks.]

Section A

1. (a) What is "Machine Learning"? What is dimension reduction in Machine Learning? Give examples. **2.75**
   (b) Describe different types of Machine learning problems with necessary examples. **3**
   (c) Describe different types of Machine learning methods with necessary examples. Make a comparison among different machine learning methods. **3**

2. (a) Define (i) False Acceptance Rate (FAR), (ii) False Rejection Rate (FRR), (iii) Equal Error Rate (EER), (iv) Receiver Operating Characteristics (ROC), (v) Success Rate or Hit Ratio, (vi) F-score, (vii) Specificity, (viii) Sensitivity, and (ix) Confusion Matrix (Please provide figures if necessary). 4.5
   (b) Discuss Overfitting vs generalization with necessary example. 2
   (c) What is "cross-validation"? Explain 4 fold cross-validation methods. 2.25

3. (a) Define: (i) input/hidden/output layers, (ii) weights, (iii) activation functions, and (iv) error 2
   (b) Define Hebb rule. Analyze the application of Hebb rule using an autoassociative memory. 3
   (c) Consider the three prototype patterns shown below: 3.75



   i) Use the Hebb rule to design a perceptron network that will recognize these three patterns p_1, p_2 and p_3.
   ii) Find the response of the network to the pattern p_t. Is the response correct?

4. (a) How would you apply a Naive Bayes classifier to characteristics that are categorical? How about features that are numerical? 2.75
   (b) Does Naive Bayes fit within the generative or discriminative classifier category? 2
   (c) Consider the following data set for Naive Bayesian classification problem. 4

Age	Income	Credit_Rating	Loan (Class)
High	Fair	Low	60,000
Low	Fair	High	70,000
Low	Excellent	High	80,000
High	Fair	High	65,000

Predict the class label for a test sample (A = Low, B = Fair, C = Low) using the Naive Bayesian classification. Note that, you may use the Laplacian correction to avoid computing probability values of zero if you want.

Section B

5. (a) Is it conceivable that during further K-Means rounds, the clustering of observations remains the same? 2
   (b) What do you understand by feature space? What is linearly separable and not separable? Explain with example. 2.75
   (c) Assume that the objective of the machine learning job is to divide the CGPA of the following students into two clusters. 3.75

A	B	C	D	E	F
3.45	2.78	2.90	3.53	2.25	3.67

We initially designate A and C as the respective centers of each cluster. To display only the final two clusters, use the k-means technique.

6. (a) Write short notes on Decision tree. 2
   (b) What is Edit Distance? Write down Edit Distance algorithm. Find out the Edit Distance between 'exercised' and 'executed'. 2.25
   (c) What is support vector? Why maximum margin in SVM technique? 2
   (d) What is the "kernel trick"? How do you compute M in terms of **W** and b? 2.5

7. (a) Demonstrate a proof of convergence for the perceptron learning rule, for a single neuron perceptron. 4

   (b) Solve the following classification problem with the perceptron rule. Apply each input vector in order, for as many repetitions as it takes to ensure that the problem is solved. Draw a graph of the problem only after you have found a solution. 4.75

$$\left\{p_1=\begin{bmatrix}2\\2\end{bmatrix},t_1=0\right\}\left\{p_2=\begin{bmatrix}1\\-2\end{bmatrix},t_2=1\right\}\left\{p_3=\begin{bmatrix}-2\\2\end{bmatrix},t_3=0\right\}\left\{p_4=\begin{bmatrix}-1\\1\end{bmatrix},t_4=1\right\}$$

Use the initial weights and bias:

$$W(0)=[0 \quad 0] \qquad b(0)=0.$$

8. A produce dealer has a warehouse that stores a variety of fruits and vegetables. When fruit is brought to the warehouse, various types of fruit may be mixed together. The dealer wants a machine that will sort the fruit according to type. There is a conveyer belt on which the fruit is loaded. This conveyer passes through a set of sensors, which measure three properties of the fruit: *shape, texture* and *weight*. These sensors are somewhat primitive. The shape sensor will output a 1 if the fruit is approximately round and a -1 if it is more elliptical. The texture sensor will output a 1 if the surface of the fruit is smooth and a -1 if it is rough. The weight sensor will output a 1 if the fruit is more than one pound and a if it is less than one pound. The three sensor outputs will then be input to a neural network. The purpose of the network is to decide which kind of fruit is on the conveyor, so that the fruit can be directed to the correct storage bin. To make the problem even simpler, let's assume that there are only two kinds of fruit on the conveyor: apples and oranges.
As each fruit passes through the sensors it can be represented by a three dimensional vector. The first element of the vector will represent shape, the second element will represent texture and the third element will represent weight as below.

$$p = \begin{bmatrix} shape \\ texture \\ weight \end{bmatrix}$$

The neural network will receive one three-dimensional input vector for each fruit on the conveyer and must make a decision as to whether the fruit is an *orange* (p_1) or an *apple* (p_2).

(a) Write down the prototype templates for *orange* (p_1) and *apple* (p_2). 1
(b) Design a perceptron to recognize these patterns. 1.5
(c) Design a Hamming network to recognize these patterns. 1.5
(d) Design a Hopfield network to recognize these patterns. 1.5
(e) Test the operation of your networks by applying several different input patterns. Discuss the advantages and disadvantages of each network. 3.25

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Total Marks: 52.5 Marks Time: 4:00 Hours

[Answer the six of the following questions Three from each Section A and Section B; Figures in the right-hand margin indicate full marks.]

Section A

1. (a) Define: 1.75
   (i) input/hidden/output layers
   (ii) weights
   (iii) activation functions
   (iv) error

   (b) Write short notes on: (i) Artificial Neurons, (ii) Artificial Neural Network. 2

   (c) Consider the following training set 3

No.	input values	desired output value
1.	$x^1 = (1, 0, 1)^T$	-1
2.	$x^2 = (0, -1, -1)^T$	1
3.	$x^3 = (-1, -0.5, -1)^T$	1

   Assume the learning rate is 0.1 and the initial weight vector is $w_0 = (1, -1, 0)^T$
   After training draw the neural network.

   (d) Make a comparison among supervised, unsupervised and hybrid ANNs in terms of learning rule, Architecture, learning algorithm and task. 2

2. (a) Draw a flow diagram to demonstrate the Genetic Algorithm process. 2

   (b) Suppose a genetic algorithm uses chromosomes of the form x = abcdefgh with a fixed length of eight gens. Each gene can be any digit between 0 and 9. Let the initial population consist of four individuals with the following chromosomes: 2

   x1 = 6 5 4 1 3 5 3 2
   x2 = 8 7 1 2 6 6 0 1
   x3 = 2 3 9 2 1 2 8 5
   x4 = 4 1 8 5 2 0 9 4

   Perform the following crossover operations:
   (i) Cross the fittest individuals (x1,x2) using one-point crossover at the middle point.
   (ii) Cross the third and fourth fittest individuals using a two-point crossover (points b and f)

   (c) Let function $f(x) = 2x - x^2/16$ be defined on the interval $[0, 31]$. Show the use of a genetic algorithm for determining the maximum of the function in the given interval. Note that the relative fitness function g(x) is defined as below: 4.75

$$g(x) = \frac{f(x)}{\sum_{x \in p^{(k)}} f(x)}$$

3. (a) Write short notes on Decision tree. 1
   (b) What is CART? Describe four important string problems in pattern recognition. 2
   (c) Write down Boyer-Moore String Matching algorithm and explain with an example. 2.25
   (d) What is Edit Distance? Write down Edit Distance algorithm and explain with an example. 2
   (e) Write short notes on: 1.5
      (i) String matching with Error
      (ii) String matching with the "Don't-Care" Symbol.

4. (a) What is support vector? Why maximum margin in SVM technique? 2
   (b) What is the "kernel trick"? How do you compute M in terms of W and b? 2.25
   (c) Write algorithm for K-means clustering algorithm. 1.5
   (d) Suppose that the machine learning task is to cluster the following students CGPA into three clusters, i.e., k = 3: 3

A	B	C	D	E	F
3.45	3.78	2.98	3.24	4.0	3.9

The distance function is Euclidean distance. Suppose initially we assign A, C, E as the center of each cluster. Use k-means algorithm to show only the final three clusters.

Section B

5. (a) Briefly describe the application of neural network. 1.25
   (b) Given a two-input neuron with the following parameters: b= 1.2, W = [3 2] and $\mathbf{p} = [-5\ \ 6]^T$, calculate the neuron output for the following functions: 1
      i. A symmetrical hard limit transfer function
      ii. A saturating linear transfer function
   (c) A single-layer neural network is to have six inputs and two outputs. The outputs are to be limited to and continuous over the range 0 to 1. What can you tell about the network architecture? Specifically: 1
      i. How many neurons are required?
      ii. What are the dimensions of the weight matrix?
   (d) Describe multiple layers of neurons with necessary figures. 2
   (e) Solve the following classification problem with the perceptron rule. Apply each input vector in order, for as many repetitions as it takes to ensure that the problem is solved. Draw a graph of the problem only after you have found a solution. 3.5

$$\left\{p_1 = \begin{bmatrix}2\\2\end{bmatrix}, t_1 = 0\right\}\left\{p_2 = \begin{bmatrix}1\\-2\end{bmatrix}, t_2 = 1\right\}\left\{p_3 = \begin{bmatrix}-2\\2\end{bmatrix}, t_3 = 0\right\}\left\{p_4 = \begin{bmatrix}-1\\1\end{bmatrix}, t_4 = 1\right\}$$

Use the initial weights and bias:

$$W(0) = [0 \quad 0] \qquad b(0) = 0.$$

6. (a) What do you understand by feature space? What is linearly separable and not separable? Explain with example. 2

(b) A produce dealer has a warehouse that stores a variety of fruits and vegetables. When fruit is brought to the warehouse, various types of fruit may be mixed together. The dealer wants a machine that will sort the fruit according to type. There is a conveyer belt on which the fruit is loaded. This conveyer passes through a set of sensors, which measure three properties of the fruit: *shape, texture* and *weight*. These sensors are somewhat primitive. The shape sensor will output a 1 if the fruit is approximately round and a -1 if it is more elliptical. The texture sensor will output a 1 if the surface of the fruit is smooth and a -1 if it is rough. The weight sensor will output a 1 if the fruit is more than one pound and a -1 if it is less than one pound. The three sensor outputs will then be input to a neural network. The purpose of the network is to decide which kind of fruit is on the conveyor, so that the fruit can be directed to the correct storage bin. To make the problem even simpler, let's assume that there are only two kinds of fruit on the conveyor: Banana and Pineapple.
As each fruit passes through the sensors it can be represented by a three dimensional vector. The first element of the vector will represent shape, the second element will represent texture and the third element will represent weight as below.

$$p = \begin{bmatrix} shape \\ texture \\ weight \end{bmatrix}$$

The neural network will receive one three-dimensional input vector for each fruit on the conveyer and must make a decision as to whether the fruit is a *banana* (p_1) or a pine*apple* (p_2).

(i) Write down the prototype templates for *banana* (p_1) or a pine*apple* (p_2). 1

(ii) Design a perceptron to recognize these patterns. 1

(iii) Test the operation of your networks by applying several different input patterns. Discuss the advantages and disadvantages of each network. 2.75

(c) Write short notes on Hamming network and Hopfield network. 2

7. (a) What do you understand about Instance-Based Learning? 1

(b) Describe the k-nearest neighbor learning. 2.75

(c) What is a distance metric? What propertles does it have to satisfy? 2
Define (any three)
(i) Minkowski-norm, (ii) Euclidean norm
(iii) Manahattan norm (iv) Hamming distance
(v) Maximum norm

(d) Write down Sequential Covering Algorithm. 2

(e) What is the difference between inductive and analytical learning methods? 1

8. (a) What is Machine learning? Describe the process of designing a learning system. 3

(b) Bayesian classifiers are statistical classifiers. Justify your answer. 1.75

(c) Why Naive Bayesian Classification is known as class conditional independence? 2

(d) What is Q learning? Describe Q learning algorithm. 2

University of Chittagong

7th Semester B.Sc (Engg.)Examination 2019

Department of Computer Science & Engineering

Course Code: CSE 715

Course Title: Machine Learning

Time: 4 Hours **Full Marks: $52\frac{1}{2}$**

[Answer six questions taking any three from each section and use separate answer script for each section]

Section A

1. a) The Apriori algorithm makes use of prior knowledge of subset support properties. 3
   i) Prove that all nonempty subsets of a frequent itemset must also be frequent.
   ii) Prove that the support of any nonempty subset s' of itemset must be at least as great as the support of s.

   b) Apriori and FP-Growth are two famous algorithms for frequent itemset generation. Compare the efficiency pf the two mining process. 1.75

   c) A database has eight transactions. Let min_sup=2. Find all frequent item sets using Aprori Algorithm. 4

TID	1	2	3	4	5	6	7	8
Items	K,C,A	A,M,D	C,K,D	K,D,A	A,C,S	D,A,S	K,S,A	A,M,K

2. a) State why Random Forest may improve the accuracy of decision tree induction. 2.75

   b) Consider the table I for a binary class problem. 6

Table I

A	B	Class Label
T	F	+
T	T	+
T	T	+
T	F	-
T	T	+
F	F	-
F	F	-
F	F	-
T	T	-
T	F	-

   i. Calculate the information gain when splitting on A and B. Which attribute would the decision tree induction algorithm choose?
   ii. Calculate the gain in the Gini index when splitting on A and B. Which attribute would the decision tree induction algorithm choose?

3. a) Bayesian classifiers are statistical classifiers. Justify your answer. 1.75
   b) Why naive Bayesian classification is known as class-conditional independence? 2
   c) Suppose playing record of tennis is shown in table II. 5

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Table II

	Outlook	Temp	Humidity	Wind	Play
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Today is Sunny, Mild, High and Strong. Using Naïve Bayesian Classifier determine if anyone will play tennis today.

4. a) It is difficult to assess classification accuracy when individual data objects may belongs to more than one class at a time. In such cases, comment on what criteria you would use to compare different classifiers modeled after the same data. 2.75

b) Suppose that we would like to select between two prediction models M1 and M2. You need to smooth both model's Recall-Precision (R-P) using interpolation between 0~1, and also need to draw R-P curve for model selection. 3

M1					
Recall	0.20	0.40	0.70	0.90	1.0
Precision	0.12	0.40	0.67	0.75	0.88

M2					
Recall	0.10	0.30	0.86	0,96	1.0
Precision	0.19	0.45	0.76	0.85	0.98

c) Construction and evaluation of a classifier require partitioning labeled data into a training set and a test set. Sketch the following methods for the following data set. 3

i) Cross Validation (k=2)

ii) Stratified Cross Validation (k=2)

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tid	Dimension1	Dimension2	Dimension3	Dimension4	Class Label
1	A1	B1	C1	D1	E1
2	A1	B1	C1	D2	E1
3	A1	B2	C1	D1	E1
4	A1	B3	C2	D1	E2
5	A2	B1	C1	D1	E1
6	A2	B3	C2	D2	E1
7	A4	B2	C1	D1	E2
8	A4	B3	C2	D1	E2
9	A4	B3	C2	D2	E1
10	A4	B2	C2	D1	E2

Section B

5. a) Suppose that the machine learning task is to cluster the following students CGPA into three clusters: 4.75

A	B	C	D	E	F
5.45	3.88	2.78	1.25	2.2	3.80

The distance function is Manhattan distance. Suppose initially we assign B, D, and F as the center/representative of each cluster, respectively. Use k-means algorithm to show only the final three clusters.

b) Use the following dissimilarity matrix to perform AGNES. Show you results by drawing a dendogram. The dendogram should clearly show the order in which the points are merged. 4

	P1	P2	P3	P4	P5
P1	0.00	0.10	0.41	0.55	0.35
P2	0.10	0.00	0.64	0.47	0.98
P3	0.41	0.64	0.00	0.44	0.85
P4	0.55	0.47	0.44	0.00	0.76
P5	0.35	0.98	0.85	0.76	0.00

6. a) How can you improve the interpretability of the learned network in backpropagation? 2
   b) Why neural network learning is referred as connectionist learning? 2
   c) Why we use a squashing function in backpropagation? 1.75
   d) For the following table, design a multilayer feed-forward neural network and show weight values after one iteration of the backpropagation algorithm, given the training instance X=(**Body Temperature=warm, Gives Birth=yes, Four-legged=no, Hibernates=yes).** Indicate your initial weight values and biases and the learning rate used. 3

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Body Temperature	Gives Birth	Four-legged	Hibernates	Class Label
Cold	no	Yes	yes	no
Cold	yes	No	no	no
Warm	no	No	no	no
Warm	no	No	yes	no
Warm	no	Yes	yes	yes

7. a) Illustrate the limitation of k-means algorithm. 3
   b) Use an example to show why k-means algorithm may not find the global optimum that is optimizing the within-cluster variation. 2.75
   c) Clustering is recognized as an important datamining task with broad applications. Give one application example that uses a clustering as a preprocessing tool for data preparation for other data mining tasks. 3

8. a) Support Vector Machine always tries to maintain maximum margin for classification. Explain it with proper sketch. 2.75
   b) What would be the best generalization for choosing the number of support vectors, if the dimensionality of data is high? 2
   c) Suppose we are given the following positively labeled data points in 2D space: {(4,1), (4,-1), (5,1), (5,-1)} and the following negatively labeled data points in 2D space: {(2,0), (0,2), (0,-2), (-2,0)}. In addition, three support vectors are {(2,0), (4,1), (4,-1),} Apply Support Vector Machine to classify data objects (6,-1). 4

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